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Cluster dynamics models of irradiation damage accumulation in ferritic iron. I. Trap mediated interstitial cluster diffusion

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The microstructure that develops under low temperature irradiation in ferritic alloys is dominated by a high density of small (2–5 nm) defects. These defects have been widely observed to move via occasional discrete hops during *in situ* thin film irradiation experiments. Cluster dynamics models are used to describe the formation of these defects as an aggregation process of smaller clusters created as primary damage. Multiple assumptions regarding the mobility of these damage features are tested in the models, both with and without explicit consideration of such irradiation induced hops. Comparison with experimental data regarding the density of these defects demonstrates the importance of including such motions in a valid model. In particular, discrete hops inform the limited dependence of defect density on irradiation temperature observed in experiments, which the model was otherwise incapable of producing. © 2015 AIP Publishing LLC.

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I. INTRODUCTION

A wide variety of detailed data on the development of intermediate dose (less than ten dpa) irradiation damage microstructures in ferritic/martensitic steels and other bcc alloys has been produced by *in situ* ion irradiation studies.^{1–7} Such experiments allow irradiation concurrent with observation in a transmission electron microscope. As with post-irradiation microscopy, *in situ* experiments can be used to quantify the damage features that form via measurements of density, size, and size distribution but with the advantage that the same area can be observed for the entire dose interval. Additionally, the dynamic behavior these experiments reveal generate quantitative and qualitative insights into the growth mechanisms, mobility, and interaction properties of visible damage features during irradiation. Such important insights are unattainable from more conventional post irradiation observation techniques.

A black dot damage microstructure is seen to dominate irradiated ferritic/martensitic alloys up to temperatures of at least 300 °C.^{1–5,8,9} At higher temperatures, extended prismatic dislocation loops with $a/2\langle 111 \rangle$ or $a\langle 100 \rangle$ Burgers vectors are clearly distinguished. A preference for $a\langle 100 \rangle$ is observed to increase with temperature, and the loops are of interstitial type.^{1–3} The black dots which form at lower temperatures are presumably also interstitial type loops of a sufficiently small size that the dislocation itself cannot be resolved. These two to five nm features become difficult to analyze because of their size, however, limiting confident assessments of type or Burgers vector. Numerous studies have indicated that these features grow in density up to a saturation value past which any further evolution is restricted to the defect size.^{4,5}

In situ experiments have provided new clues as to the properties of these defects. Under both electron and heavy ion irradiation, the thermally immobile damage features have been widely observed to undergo occasional but rapid

translations spanning distances on the order of a few to a few tens of nanometers which punctuate extended periods where no motion occurs at all.^{2,7,9–14} Notably, these “hops” generally occur only when the ion or electron beam is operating and occur over the entire regime of black dot damage, even at temperatures as low as 50 K. The limited mobility of such features contrasts vividly with predictions about their motion based on molecular dynamics simulations, in which interstitial clusters diffuse along one dimensional trajectories with very low (less than 0.1 eV) activation energies.^{15–19} The reconciliation of disparate expected and observed behaviors has been rationalized by “traps,” broadly defined as a variety of impurities or impurity defect complexes which bind to the dislocation loops and lower their mobility. Though multiple candidates for the composition of such traps have been proposed, these remain to be conclusively identified; however, trapping has been invoked to explain the lower than expected thermal mobility of loops²⁰ as well as the hops seen in irradiation environments.^{11,12}

The kinetic properties of interstitial clusters revealed through *in situ* experiments have been applied erratically in various modeling approaches, particularly cluster dynamics (CD) models, which have sought to characterize the formation of black dot damage with varied degrees of apparent success.^{21–24} This work seeks to establish a more systematic approach to incorporate the known aspects of interstitial cluster behavior into rate theory based models, and demonstrates that the resultant description of observable damage feature evolution is improved over prior modeling attempts.

II. METHODS

The cluster dynamics method is an expansion of the classical rate theory approach to quantifying the effects of radiation, which operates by integrating a coupled set of reaction-diffusion equations

$$\frac{dC_i}{dt} = \nabla(D_i \nabla C_i) - D_i k_i^2 C_i + R_i(C) + g_i, \quad (1)$$

which represent the concentration C_i of each defect or impurity species i . Each equation contains a unique sink strength for depletion at pre-existing sinks k_i^2 and rate of production in damage events g_i . In our particular model, the only sinks considered are network dislocations which are treated using the cylindrical cell approximation,²⁵ such that the sink strength is

$$k_i^2 = \frac{2\pi\rho_d}{\log(\pi\rho_d r_i^2)^{-1/2}} \beta_i \quad (2)$$

for a dislocation density of ρ_d . The dislocation bias β is applied only to absorption of interstitial type defects. The equations are coupled through the reaction terms $R_i(C)$ which are constituted from a potentially very large set of individual mass action kinetics problems arising from defect cluster interactions of the form $A + B \rightarrow C$. All possible reactions producing or consuming a given species i must be considered, such that

$$R_i(C) = \sum_{n+m \rightarrow i} k_{n,m}^+ C_n C_m - \sum_j k_{i,j}^+ C_i C_j, \quad (3)$$

where the reaction rate coefficients $k_{i,j}^+$ are determined in a manner consistent with the methodology for k_i^2 . It is common to assume that these coefficients take the Smoluchowski rate

$$k_{i,j}^+ = 4\pi r_{ij}(D_i + D_j)\beta_{ij}, \quad (4)$$

where the r_{ij} is the radius of the reaction volume. This distance is taken to be the sum of the radii of the reacting clusters and some interaction distance such that $r_{ij} = r_i + r_j + r_0$. The bias β is applied only if both defects are interstitial type. Assumptions of spherical symmetry and three dimensional diffusion are embedded in these coefficients, a potential source of error that will be revisited later. Dissociation events are also included in the construction of the reaction terms, but their relevance to this study is marginal and the interested reader is referred elsewhere.²¹

The diffusion coefficient features prominently in the reaction, sink loss, and spatial transport of cluster species. As such, the fidelity of the cluster diffusivity parameters is of paramount importance to the success of these models. Indeed, embedding the physics of interstitial cluster transport into the rate equations constitutes the focus of this work. Typically, information about cluster migration is expressed in terms of Arrhenius relationships governed by

$$D_i = D_{0,i} \exp \frac{-E_{m,i}}{k_b T}, \quad (5)$$

where $D_{0,i}$ is a diffusion prefactor and $E_{m,i}$ is the migration energy for the cluster in question. A number of CD and object kinetic Monte Carlo (OKMC) studies have pondered the proper choice of E_m to accommodate the trap mediated nature of interstitial cluster diffusion.^{21,26,27} We review

some of these past choices, and examine the implications they have for the development of the black dot microstructure at various temperatures.

The most common approach extends the low thermal mobility of visible clusters to arbitrarily small crowdion bundles, in some cases even treating these clusters as though they were sessile. Other researchers have imposed a crowdion mobility that varies from computational predictions at small size to experimental values at larger sizes,²¹ a possibility we examine as well. Table I contains four sets of migration energy parameterizations which are used in this cluster dynamics model to examine the implications of the mobility of small interstitial clusters. The first two sets represent interstitial clusters being bound to trapping sites, regardless of size, while the other sets allow free glide for small clusters in accordance with prior studies. The long range migration of vacancies and vacancy clusters in iron is also reduced by binding with impurities, particularly carbon.^{28–30} Consequently, though *ab initio* values for vacancy cluster migration are examined in set II, higher effective activation energies are assumed in all other cases.

We add to these sets of defect cluster mobility an additional effect which accommodates the dynamic behavior observed during *in situ* experiments where thermally immobile loops undergo erratic motion. In such a trap mediated diffusion environment, the ultimate mobility of loops is governed by a series of ballistic detrapping events induced by energetic recoils. Following each of these events, a crowdion bundle glides with low activation energies until encountering a new trapping site. The length and frequency of these apparent hops are determined by the trap density and irradiation condition, respectively.

The effective diffusivity that results from a random walk of the irradiation induced motions can be described according to

$$D_i^{dt} = \nu_i^{dt} \frac{\lambda_i^2}{2N}, \quad (6)$$

TABLE I. Migration energy parameter sets for point defects and defect clusters as implemented in the cluster dynamics model. The notation $A \rightarrow B$ indicates a linear increase in migration energy between A and B over the range of clusters specified.

Cluster	D_0 (nm ² /s)	E_m (eV)			
		I	II	III	IV
v_{5+}	0	...	...	...	...
v_4	$D_{0,1/4}$	1.3	0.48 (Ref. 31)	1.3	1.3
v_3	$D_{0,1/3}$	1.3	0.35 (Ref. 31)	1.3	1.3
v_2	$D_{0,1/2}$	1.3	0.62 (Ref. 31)	1.3	1.3
v_1	10^{11}	1.3	0.67 (Ref. 31)	1.3	1.3
i_1	10^{11}	0.34 (Ref. 32)	0.34	0.34	0.34
i_2	$D_{0,1/2}$	0.42 (Ref. 32)	0.42	0.42	0.42
i_3	$D_{0,1/3}$	0.43 (Ref. 31)	0.43	0.43	0.43
$i_4 - i_6$	$D_{0,1/n}$	1.3	1.3	0.1	0.1
$i_7 - i_{19}$	$D_{0,1} \times n^{-0.7}$	1.3	1.3	0.1 $\rightarrow$ 1.3	0.1 $\rightarrow$ 0.55
$i_{20} - i_{37}$	$D_{0,1} \times n^{-0.7}$	1.3	1.3	1.3	0.55 $\rightarrow$ 1.3
$i_{37} +$	$D_{0,1} \times n^{-0.7}$	1.3	1.3	1.3	1.3

where the activation frequency ν and the discrete hop length λ are associated with jumps between trapping sites, rather than the more familiar jumps between lattice sites typically employed to construct the diffusivity of lattice defects. The overall dimensionality of these motions is assumed to be three dimensional ($N=3$), as this remains embedded in the reaction rate constants. In electron irradiation, the frequency of hops has been studied fairly comprehensively, providing a clear linear relationship with the dose rate and a proportionality constant near unity.¹⁰ The large number of atoms involved in cascades—though not necessarily displaced in them—suggests that the detrapping frequency may be significantly higher in such environments. Though no equivalently rigorous study exists, initial assessments of detrapping in the heavy ion environment support this view.⁹

This model equates the hop length with the mean free path between trapping interactions. As crowdion motion is nominally one dimensional, hop length becomes a simple function of the trap density and interaction cross section. Experimental literature has provided some evidence that the dominant traps are interstitial impurity or solute atoms which act by obstructing the passage of dislocations.^{11,12} In these studies, the interaction cross section scales with loop size according to $\sigma(r) \propto r$. Consequently, the hop length becomes

$$\lambda_i = (4\pi r_i r_T \rho_T)^{-1}, \quad (7)$$

where r_T is and interaction distance between the dislocation loop and the trap, and ρ_T is the trap density.

While the activation frequencies for these motions are known, or can be estimated to first order, the trap density presumably varies widely with material composition. Additionally, the clusters produced as primary damage vary based on the irradiation condition. This leaves a large phase space of potential parametrization to explore. This study is largely confined to two combinations of irradiation condition and assumed material properties designed to directly correspond to *in situ* experiments for which data have been reported at temperatures ranging from 50 to 650 K. The first of these is an electron irradiation experiment in high purity iron as reported by Arakawa *et al.*¹³ The second is a series of heavy ion irradiations performed on a variety of iron chrome based commercial and model alloys.^{4,5,9}

To emulate electron irradiation damage conditions, primary damage was introduced in the model as Frenkel pairs with a uniform distribution with respect to depth. In the ion irradiation case, damage was divided amongst clusters of various size according to the procedure outlined by Xu *et al.*²¹ The relative magnitudes of these source terms is shown in Figure 1 for a 1 MeV Kr beam striking a 100 nm thick iron target at a 15 degree angle of incidence. The depth dependence of source term magnitudes in the model matched the SRIM full cascade distribution of damage energy according to the method suggested in Stoller *et al.*³³ The nominally higher purity of the material studied in the electron irradiation case was reflected by a lower magnitude for the trap density in these materials. The parameters used in each case are shown in Table II.

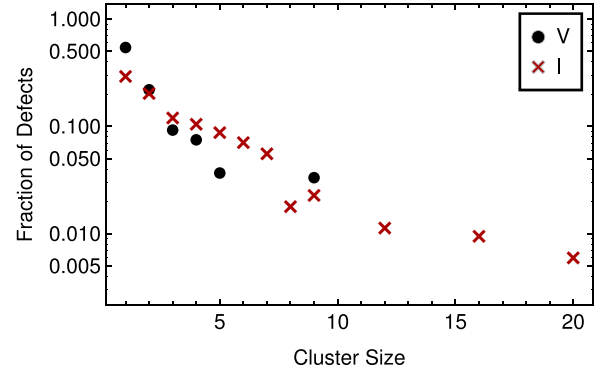


FIG. 1. Fraction of total defects contained in clusters of a given size for vacancy (V) and interstitial (I) defects in the ion irradiation environment.

III. RESULTS

A. Thermal diffusion

To examine the effect of temperature, calculations were performed at a variety of irradiation temperatures between 150 and 600 K. Such calculations were done for each of the two experimental conditions and both with and without the added mobility of beam induced detrapping. In all cases, each of the four interstitial mobility sets was tested. Figure 2 demonstrates the visible defect density at which saturation was reached as a function of temperature when no detrapping terms were included. Absent data points indicate that saturation was not reached within the dose increment simulated, or occurred at unphysical magnitudes.

Frenkel pair damage generated similar defect behavior irrespective of defect mobility assumptions, demonstrating a variation of defect density exceeding four orders of magnitude within the span of about one hundred degrees. Though increasing the number of mobile interstitials shifted the relative magnitudes of this trend, it did not alter the behavior itself in any way, and changes in vacancy mobility had no effect whatsoever. In models emulating ion irradiation, mobility set IV behaved similarly to the electron irradiation calculations, but the other sets all exhibited an even stronger temperature dependence, with a failure to reach saturation at temperatures below 400 K. The effect of vacancy mobility in the ion irradiation case was barely discernible, consistent with the Frenkel pair case.

TABLE II. Simulation parameters for cluster dynamics calculations.

Parameter	Electron	Ion
Trap density, ρ_T (ppm)	10	200
Dose rate (dpa/s)	5×10^{-4}	8.6×10^{-4}
Hop activation, ν^{dt} (dpa ⁻¹)	1	30
Burgers vector, b (nm)		0.249
Trap radius, r_T		$1.5b$
Interaction Distance, r_0		$1.5b$
I cluster radius $r_I(n)$		$(\frac{\Omega n}{4\pi})^{1/2}$
V cluster radius $r_V(n)$		$(\frac{3\Omega n}{4\pi})^{1/3}$
Dislocation Bias β		1.1
Dislocation Density ρ_d (nm ⁻²)		2.5×10^{-4}

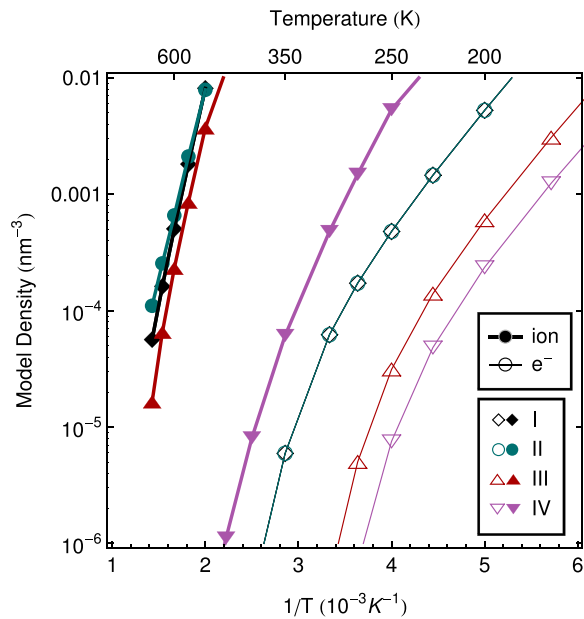


FIG. 2. Saturation density of interstitial clusters larger than 2.5 nm in diameter as a function of temperature for various cluster mobilities without beam activated motion. Open data points indicate electron irradiation simulations, while closed points indicate ion irradiation. Cases where saturation was not reached by 10 dpa or exceeded the displayed range were omitted.

B. Trap mediation effects

Figure 3 demonstrates the temperature dependence when beam activated hops are incorporated into the reaction-diffusion network. In stark contrast to the previous calculations, sets I–III all give rise to a predominantly athermal saturation density. Set IV also produces athermal behavior at cryogenic temperatures, but diverges sharply from the other calculations above room temperature to yield densities consistent with its saturation behavior in the absence of beam induced detrapping. Below this temperature, the migration energies assigned to the larger clusters do not influence saturation behavior at all, and the defect evolution predicted in

the different cases is indistinguishable. When detrapping is considered, vacancy mobility begins to impact the defect population above room temperature, although the density of loops is not altered by more than a factor of 2. Above 550 K, the observed loop density begins to drop off sharply for sets I–III, consistent with the high temperature behavior in the thermal case.

The experimental observations that motivate the development of these models are also shown in Figure 3. These include an Fe-12Cr model alloy⁵ and NF616,^{4,9} in which the quantities reported were normalized by the authors to a reference value of the defect density at 50 K in all cases. These normalizations are performed because the density of defects as observed is not equivalent to the real density of defects in the material. The visibility criterion renders a large fraction of defects undetectable, and is responsible for the bulk of the difference. Of the nominally visible defects, some unknown fraction will inevitably be obscured by one another or other features in the microstructure. Others still may not be clearly distinguishable as defects, particularly those with sizes near the visibility threshold. These factors are largely systematic, however, and normalization across equivalently performed experiments allows them to be eliminated.

The calculations for electron irradiation using differing thermal mobility set were also nearly indistinguishable at sufficiently low temperatures, as shown in Figure 4. The influence of temperature was slightly more pronounced in the cryogenic region when compared to the heavy ion cases. Additionally, the point of departure from athermal behavior occurred for all mobility sets very near to 300 K. As in the case of ion irradiation, the trends in defect density at higher temperature mimicked the thermal calculations where strong temperature dependence occurred. A defect density which sharply transitions from athermal behavior to thermal dependence is consistent with the observations of Arakawa *et al.*¹³ The values reported from these experiments are also shown in Figure 4 as a point of comparison, normalized to emphasize

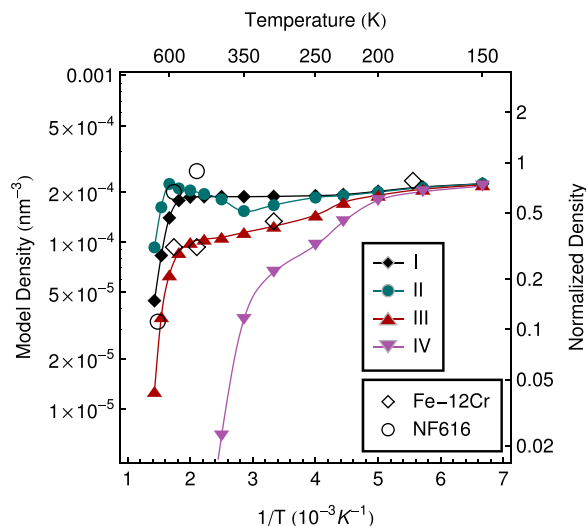


FIG. 3. Calculated saturation density for ion irradiation with beam activated motion. Experimental data are shown for purposes of comparison, as normalized to the defect density at 50 K.

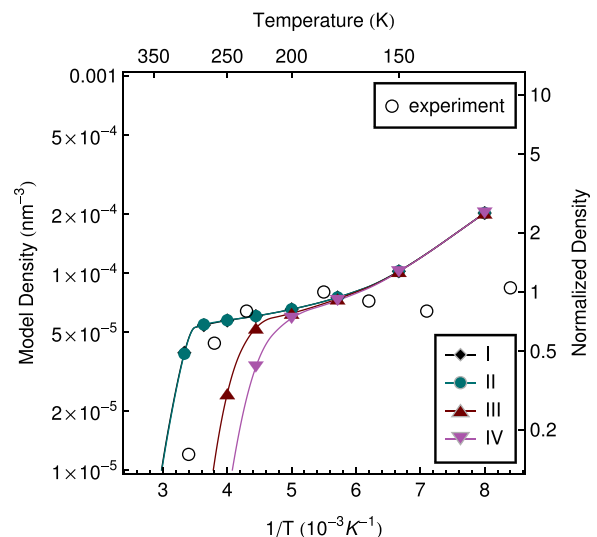


FIG. 4. Calculated saturation density for electron irradiation with beam activated motion. Experimental data are shown normalized to the lowest temperature value.

the temperature dependence. The temperature at which this transition occurs in the experimental data is roughly equivalent to the model calculations, though the exact value differs slightly with respect to assumptions about the thermal mobility of interstitials.

C. Evolution with dose

In addition to a drastically altered temperature dependence, the inclusion of irradiation activated kinetic pathways strongly modified the evolution of the defect population with respect to dose. In the ion irradiation model, several dpa elapsed before defect density saturated at high temperatures, and saturation did not occur at lower temperatures. By contrast, when detrapping effects were considered, a peak defect density was reached well within 1 dpa in all cases. Furthermore, the density declined for the remainder of the simulation after reaching its peak value in a number of cases. This phenomenon was observed in electron irradiation models as well, with peak densities occurring at doses typically about one order of magnitude lower than those seen in the ion irradiation models.

The decline in density past the saturation dose responded strongly to vacancy mobility. Using migration sets I, III, and IV, the evolution of density with respect to dose followed a predictable pattern. Though the magnitude of the peak density varied, the subsequent decline in the population spanned a factor of two to three at low temperatures. This pattern extended up to the point where the peak density begins to follow the thermal case. This consistency was particularly evident with set I, where the magnitude of the peak density was also unchanged over the relevant temperature range. The effects of vacancy mobility were more pronounced, as demonstrated in Figure 5 which shows the evolution of damage for set II, where vacancies begin to exhibit thermal mobility near room temperature.

Below room temperature, the population follows similar trends as the other mobility sets. However, as the temperature

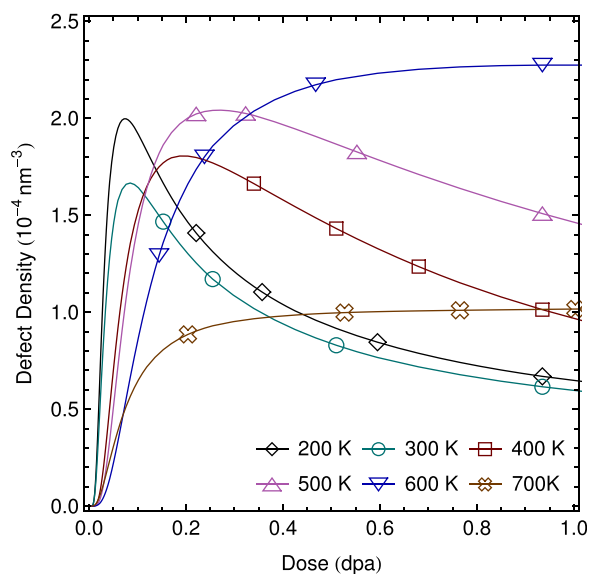


FIG. 5. The density of defects larger than 2.5 nm in diameter throughout the first dpa of ion irradiation using parameter set II.

increases, the peak density begins to shift to higher doses. This effect was not seen with the other mobility sets. In addition to delaying saturation, increasing temperature also reduces the subsequent decline of the loop population in set II. Above 600 K, a discernible peak is not present in the evolution of the population, rather the density approaches a steady state value.

Another key characteristic of the defect population which continues to evolve after the saturation point is the average cluster size. Notable growth of visible clusters was not observed for any of the thermal models, instead the mean size was sharply confined to the visibility threshold throughout the simulations. Growth did occur in the trap mediated system, often to surprising extents. As with density, the rate of growth exhibited an extended regime of temperature independence. Set II demonstrated slightly different behavior, and the rate of growth decreased with temperature. Figure 6 illustrates the difference in growth behavior between sets I and II over the first dpa of ion irradiation. Notably, the temperature range over which growth slows roughly coincides with the range over which the saturation dose becomes increasingly belated.

IV. DISCUSSION

A. Loop density evolution

The defect populations that develop in each calculation can be understood in terms of stable cluster nucleation. A typical nucleation and growth problem manifests when damage arrives as Frenkel pairs. A chain of at least three reactions is required to produce a stable nucleus, and as the density of nuclei increases an ever larger portion of the primary damage contributes to growth by absorbing at the existing clusters and is unavailable for further nucleation. When interstitial clusters are generated directly in cascades,

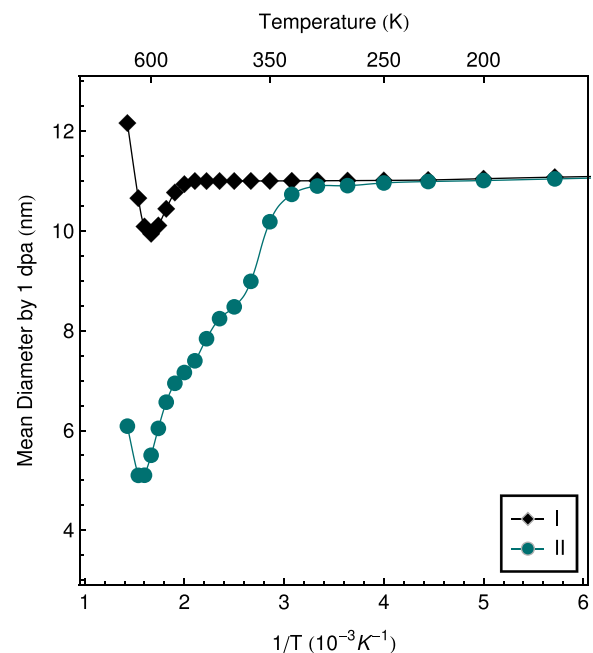


FIG. 6. Average diameter of visible loops following simulated ion irradiation to a dose of 1 dpa, as a function of temperature.

however, no such transition can occur. The rate of nucleation in the model becomes independent of the state of the system, and the density of nucleation sites in the system grows without limit. This nucleation catastrophe dominates except at sufficiently high temperatures such that the nuclei themselves become mobile, allowing coalescence events which limit the cluster density.

Introducing discrete hops as a mechanism for the diffusion of nuclei offers an alternate pathway through which the nuclei can coalesce, reach sinks and surfaces, or shrink by interaction with the vacancy population. In effect, this places intrinsic limits on the nucleus density. This informs the characteristically athermal saturation density the model predicts below roughly 550 K for cascade damage and 250 K for Frenkel pair damage. Above these temperatures, a strong temperature dependence is observed in the experiments. In our model, this dependence is driven by thermal mobility of directly implanted nuclei in the cascade damage scenario, and the thermal kinetics of loop nucleation in the Frenkel pair case. Notably, the 550 K transition point coincides with the temperature at which the larger-sized loop population begins to become increasingly dominated by $\langle 100 \rangle$ loops in cascade damage conditions. The absence of $\langle 111 \rangle$ loops at these temperatures may be a simple consequence of the thermal depletion of $\langle 111 \rangle$ nuclei.

The declining density of visible loops might have a number of possible explanations. The simplest might be loss of loops at the free surfaces, an effect often observed in *in situ* experiments. To this, we add alternative processes such as the coarsening of loops via mutual interactions enabled by the hopping process or reaction of mobile loops with an increasing density of retained vacancies and small vacancy clusters. These three options are difficult to assess quantitatively. Though surface loss plays some role, the net interstitial content only differs from the net vacancy content by a factor of about 30%, and this difference in content is established well before saturation. This is consistent with an expectation that the clusters with the longest trap spacings are most likely to escape, and these are sub-visible in size. Further, the density decline is not universal as would be expected if surface losses were solely responsible, and does not occur in simulations where vacancies have high mobility. We conclude that the loss of loops to surfaces does not act alone to drive the reduction in visible population.

Coarsening is a much stronger candidate to explain this phenomenon. The rate at which mutual interactions proceed in the model can be estimated by applying the reaction rate from Eq. (4) to the average populations. When all visible loops are represented in terms of a single average size, the mutual interaction rates between visible clusters are of the order of several events per loop per dpa, declining as the average size grows. As such, a coarsening mechanism certainly contributes, and is consistent with a declining rate of loop loss as the simulation progresses and loops become larger. That loops are lost more slowly, or not at all, when vacancies are mobile is less straightforward to explain. It stands to reason, however, that in this environment, new loops can be continuously nucleated, as the existing loops remain small and the background vacancy density remains low in such

cases. In scenarios where the loops grow quickly and the vacancies are frozen in, newly produced small interstitial clusters annihilate on vacancies or absorb on the large clusters rather than grow to visible size.

Stable vacancies suppressing new loop formation by way of annihilation are related to the third possible loss mechanism that existing loops absorb vacancies which demotes them to below visible sizes. The slowing rate of decline with dose is also consistent with this interpretation, as at earlier doses a larger fraction of the loops are sized at or near the visibility threshold. At higher doses when the mean size is larger, the broadening of the size distribution by vacancy absorption pushes fewer visible clusters to sub-visible sizes. Although the greater number of reactions required to render a visible loop invisible suggests that coarsening or surface loss is the dominant process, the change in loss behavior prompted by highly mobile vacancies points to vacancy interactions as the controlling factor. It appears that a complex interdependence of these processes contributes to the loss of visible damage, and that no mechanism operates alone.

The coarsening of visibly sized interstitial clusters appears to be an inevitable consequence of the motion of interstitial clusters of arbitrary size. This coarsening process occurs at all temperatures below the point where loop migration becomes thermally activated. Above these temperatures, fewer visible loops develop, and though the interaction rates are lower, the observation of coalescence processes would still be expected according to this model. In a real microstructure, extended loops are typically not observed at or below room temperature in ion irradiation, and the visible damage typically remains in the form of black dots independent of dose. These defects are observed to organize into rafts or loop strings, however, a phenomenon which is observed following either neutron or ion irradiation near room temperature.^{1,2,34,35} Though not explicitly considered as rafts, the large features which develop in the trap mediation model might best be interpreted in such a manner.

B. Comparison to other models

The temperature dependence of defect populations reported by experimental data—and the similar behavior of these trap mediation based models—suggests that long-range, thermally activated diffusion is not the driving force for microstructural evolution below the aforementioned critical temperatures. Any model designed with simple thermal diffusion mechanisms in mind will inevitably fail there. Indeed, many past cluster dynamics based studies have encountered difficulties constructing a model compatible with cascade damage conditions. We point to two studies in particular, one examining defect formation in iron,²² and the other in molybdenum.²¹ In both cases, the authors attack the problem with a high fidelity, multi-scale model that incorporates a state of the art representation of primary damage, and find defect densities that are (much) too high.

In the former study (in iron), the authors address this happenstance by rejecting the original, physically based set of source terms to build a model which considers far fewer interstitial clusters than MD simulations would suggest. The

justification they provide for this alteration demands that interstitial clusters preferentially absorb vacancies during the later stages of cascade healing. There is little in the way of evidence to support the idea that early time cascade evolution follows such a trend. Quite the opposite, it would invalidate the basis of the so-called production bias model, which is predicated on a significant fraction of interstitials leaving cascades in the form of one-dimensionally mobile clusters.

The molybdenum study by Xu *et al.* took a different approach to resolve the same problem. Instead of assuming that the larger interstitial clusters are fully immobile, they allowed the mobility of smaller interstitial clusters to vary with size, motivating the inclusion of mobility sets III and IV in this study. With this “optimized” mobility, the authors were able to reproduce the defect evolution in Mo at 350 K across two orders of magnitude in dose rate, but the fit was only presented for a single temperature. As demonstrated by set IV in our results, such an approach retains a strong temperature sensitivity in the defect evolution. As the migration energies for interstitials in such an optimization approach are chosen without any other physical basis, they ought to, at least, demonstrate an ability to follow experimental trends across some non-trivial range of temperatures or damage sources. This condition apparently cannot be met for the experimental data in iron.

The trap mediation model presented here essentially addresses the same discrepancy between modeling predictions and observed results. It differs from other reconciliation attempts primarily in that it is based on a well-documented physical phenomenon rather than *ad hoc* parameter adjustment. The data presented here do not exclude other possible mechanisms for facilitating cluster interaction—the direct growth or shrinkage of clusters in cascade events for instance—but it does indicate that some athermal mechanism in fact dominates the microstructural evolution below the aforementioned temperatures. The discrete hops observed under irradiation are a strong candidate for that mechanism, as demonstrated by these calculations.

The authors note that the effective mobility afforded to interstitial clusters in this approach is inherently compatible with models which rely on the one dimensional motion of crowdions, whereas common parametrizations in the literature which hold these clusters entirely immobile are not. Furthermore, a defect formation mechanism based on detrapping in irradiation events is consistent with long-standing observations of defect accumulation in ion irradiated iron which suggest that visible clusters are not formed until doses at which cascades have overlapped spatially.^{7,36,37} Though these modeling results indicate that the trapping and detrapping of interstitial clusters are consistent with these observations and contribute significantly to microstructural evolution, we do not suggest that this is the only active athermal mechanism. The cascade collapse process is impacted by pre-existing damage which modifies the nature of the produced cluster distribution³⁸ and this will likely act in concert with detrapping to influence microstructural evolution, particularly at temperatures where vacancies are immobile.

The magnitudes of the defect population presented in this paper are admittedly higher than the range of values

typically reported from experimental measurements. We have argued that the trends with respect to temperature are of greater importance to identifying the crucial aspects of defect physics. These differences in magnitude could be easily resolved by parameter adjustment, for instance by tuning the trap density or detrapping frequency; however, this process was deliberately avoided. The nature of the traps themselves is shrouded in uncertainty, and certain features of the defect population might be strongly impacted by differing assumptions about the traps. For example, if impurity-defect complexes are responsible for the trapping effects, the density of traps would take some non-trivial dose interval to become fully populated and the dose at which peak visible damage is reached would be noticeably delayed.

The evolution of the defect population in this model below the transition temperatures is governed strongly, though not entirely, by the mobility of larger interstitial clusters. For this reason, our model is incomplete, as these clusters diffuse one dimensionally. Precisely, because this aspect of the physics is not considered in this study, more detailed comparisons of the defect population with experiments have been avoided. Clusters which diffuse one dimensionally will preferentially absorb on large cross section sinks, potentially impacting the eventual growth rates and size distributions of the visible clusters. This will certainly affect the coarsening processes and long term defect evolution which arise in the trap mediated model, and may influence the nucleation densities as well. This possibility is addressed in Paper II, which analyzes the impact of reaction kinetics.³⁹

V. CONCLUSIONS

The defects observed in ferritic alloys following low temperature irradiation do not arise through thermally activated long range diffusion. Defect migration energies spanning the range of conceivable parameters fail to reproduce experimentally observed trends in defect density below 550 K in cascade damage conditions and 300 K in Frenkel pair damage conditions. The trap mediated nature of interstitial mobility requires a more complex treatment than parametrization with a single effective activation energy can provide. An athermal mechanism for the formation of visible damage is apparently necessary to match the temperature dependence of damage formation in these temperature ranges. An alternative model which accounts for irradiation activated hops for prismatic loops and crowdion bundles produces defect densities in closer accordance with experimental data, and indicates that these motions may play a key role in the development of irradiation microstructures at low temperature.

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- ¹M. Hernandez-Mayoral, Z. Yao, M. L. Jenkins, and M. A. Kirk, *Philos. Mag.* **88**, 2881 (2008).
- ²Z. Yao, M. Hernandez-Mayoral, M. Jenkins, and M. Kirk, *Philos. Mag.* **88**, 2851 (2008).
- ³Z. Yao, M. Jenkins, M. Hernández-Mayoral, and M. Kirk, *Philos. Mag.* **90**, 4623 (2010).
- ⁴C. Topbasi, A. T. Motta, and M. A. Kirk, *J. Nucl. Mater.* **425**, 48 (2012).
- ⁵D. Kaoumi, J. Adamson, and M. Kirk, *J. Nucl. Mater.* **445**, 12 (2014).
- ⁶M. A. Kirk, P. M. Baldo, A. C. Liu, E. A. Ryan, R. C. Birtcher, Z. Yao, S. Xu, M. L. Jenkins, M. Hernandez-Mayoral, D. Kaoumi, and A. T. Motta, *Microsc. Res. Tech.* **72**, 182 (2009).
- ⁷M. Jenkins, Z. Yao, M. Hernández-Mayoral, and M. Kirk, *J. Nucl. Mater.* **389**, 197 (2009).
- ⁸M. L. Jenkins, C. A. English, and B. L. Eyre, *Philos. Mag. A* **38**, 97 (1978).
- ⁹C. Topbasi, "Microstructural evolution of ferritic-martensitic steels under heavy ion irradiation," Ph.D. thesis (Pennsylvania State University, 2014).
- ¹⁰Y. Satoh, H. Matsui, and T. Hamaoka, *Phys. Rev. B* **77**, 094135 (2008).
- ¹¹Y. Satoh and H. Matsui, *Philos. Mag.* **89**, 1489 (2009).
- ¹²T. Hamaoka, Y. Satoh, and H. Matsui, *J. Nucl. Mater.* **433**, 180 (2013).
- ¹³K. Arakawa, H. Mori, and K. Ono, *J. Nucl. Mater.* **307–311**, 272 (2002).
- ¹⁴K. Arakawa, M. Hatanaka, H. Mori, and K. Ono, *J. Nucl. Mater.* **329–333**, 1194 (2004).
- ¹⁵B. Wirth, G. Odette, D. Maroudas, and G. Lucas, *J. Nucl. Mater.* **244**, 185 (1997).
- ¹⁶B. Wirth, G. Odette, D. Maroudas, and G. Lucas, *J. Nucl. Mater.* **276**, 33 (2000).
- ¹⁷Y. Osetsky, D. Bacon, A. Serra, B. Singh, and S. Golubov, *J. Nucl. Mater.* **276**, 65 (2000).
- ¹⁸Y. N. Osetsky, D. J. Bacon, A. Serra, B. N. Singh, and S. I. Golubov, *Philos. Mag.* **83**, 61 (2003).
- ¹⁹N. Soneda and T. D. de La Rubia, *Philos. Mag. A* **81**, 331 (2001).
- ²⁰K. Arakawa, K. Ono, M. Isshiki, K. Mimura, M. Uchikoshi, and H. Mori, *Science* **318**, 956 (2007).
- ²¹D. Xu, B. D. Wirth, M. Li, and M. A. Kirk, *Acta Mat.* **60**, 4286 (2012).
- ²²E. Meslin, A. Barbu, L. Boulanger, B. Radigue, P. Pareige, K. Arakawa, and C. Fu, *J. Nucl. Mater.* **382**, 190 (2008).
- ²³A. H. Duparc, C. Moingeon, N. S. de Grande, and A. Barbu, *J. Nucl. Mater.* **302**, 143 (2002).
- ²⁴J. Marian and V. V. Bulatov, *J. Nucl. Mater.* **415**, 84 (2011).
- ²⁵H. Trinkaus, V. Naundorf, B. Singh, and C. Woo, *J. Nucl. Mater.* **210**, 244 (1994).
- ²⁶C. Becquart, A. Barbu, J. Bocquet, M. Caturla, C. Domain, C.-C. Fu, S. Golubov, M. Hou, L. Malerba, C. Ortiz, A. Souidi, and R. Stoller, *J. Nucl. Mater.* **406**, 39 (2010).
- ²⁷C. Domain, C. Becquart, and L. Malerba, *J. Nucl. Mater.* **335**, 121 (2004).
- ²⁸C. Domain, C. S. Becquart, and J. Foct, *Phys. Rev. B* **69**, 144112 (2004).
- ²⁹D. Terentyev, N. Anento, A. Serra, V. Jansson, H. Khater, and G. Bonny, *J. Nucl. Mater.* **408**, 272 (2011).
- ³⁰T. Jourdan, C. C. Fu, L. Joly, J. L. Bocquet, M. J. Caturla, and F. Willaime, *Phys. Scr.* **T145**, 014049 (2011).
- ³¹C. C. Fu, J. D. Torre, F. Willaime, J. L. Bocquet, and A. Barbu, *Nat. Mater.* **4**, 68 (2005).
- ³²C. C. Fu, F. Willaime, and P. Ordejon, *Phys. Rev. Lett.* **92**, 175503 (2004).
- ³³R. Stoller, M. Toloczko, G. Was, A. Certain, S. Dwaraknath, and F. Garner, *Nucl. Instrum. Methods Phys. Res. B* **310**, 75 (2013).
- ³⁴M. Eldrup, B. Singh, S. Zinkle, T. Byun, and K. Farrell, *J. Nucl. Mater.* **307–311**, 912 (2002).
- ³⁵S. Zinkle and B. Singh, *J. Nucl. Mater.* **351**, 269 (2006).
- ³⁶I. Robertson, M. Kirk, and W. King, *Scr. Metall.* **18**, 317 (1984).
- ³⁷M. Kirk, I. Robertson, M. Jenkins, T. Black, and J. Vetrano, *J. Nucl. Mater.* **149**, 21 (1987).
- ³⁸R. E. Stoller and S. G. Guiriec, *J. Nucl. Mater.* **329–333**, 1238 (2004).
- ³⁹A. Kohnert and B. Wirth, *J. Appl. Phys.* **117**, 154306 (2015).